MATH 101 Unit 1: Expressions, Equations, and Inequalities

Practice 2 | Version 202610122020.35 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

QUESTIONS	ANSWERS
1. Simplify. In (a), x and y are nonzero; give positive exponents. (a) $\frac{6x^{-3}y^{3}}{3xy^{-1}}$ (b) $\sqrt{48}$	1. (a) 10 pts (b) 10 pts
2. Simplify and state all values excluded by the original expression. $\frac{x^{2}-9}{x^{2}+2x-15}$	2. Expression 14 pts Excluded values 6 pts
3. (a) Expand and simplify. (b) Factor completely. (a) $(3x-3)(x+3)$ (b) $2x^{2}-6x-8$	3. (a) 10 pts (b) 10 pts
4. Solve both equations. In (b), state original restrictions and check your answer. (a) $x^{2}+2x-8=0$ (b) $\frac{5}{2x-6}+\frac{1}{x-3}=\frac{7}{2}$	4. (a) 10 pts (b), with check 10 pts
5. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 4$	5. Derived bounds 12 pts Interval 8 pts

Instructor answer key: MATH 101 Unit 1: Expressions, Equations, and Inequalities

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

$\frac{2y^4}{x^4}$ — 10 pts

(b)

$4\sqrt{3}$ — 10 pts

2. Expression

$\frac{x+3}{x+5}$ — 14 pts

Excluded values

$\{-5, 3\}$ — 6 pts

3. (a)

$3x^2 + 6x - 9$ — 10 pts

(b)

$2(x-4)(x+1)$ — 10 pts

4. (a)

$x = -4,\ 2$ — 10 pts

(b), with check

$x = 4;\ x \neq 3$ — 10 pts

5. Derived bounds

$-\frac{3}{2} \leq x \leq \frac{5}{2}$ — 12 pts

Interval

$[-\frac{3}{2}, \frac{5}{2}]$ — 8 pts